Arunabha Majumder

Design Portfolio | Three hardware projects

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Each project shows my role, the CAD work and one design decision made for cost, quality or manufacturability. No confidential or export-controlled material is included.

1. HYBRID-ACTUATED SHOULDER EXOSKELETON

PhD, Aalborg University, 2023 - present | *tested prototype*

- **My role:** Concept, sketches, mechanical design, prototyping, sensor integration and testing.
- **CAD work:** SolidWorks parts and assembly of the joint, frame and actuator mounts, updated after each build.
- **Design decision:** Used a parallel spring to carry most of the gravity load, so a small 6 Nm motor replaced a larger one: lower energy, lower mass and a cheaper, more compact actuator. Parts were 3D printed so each test round fed straight into the next version.
- **Result:** More than 25% lower energy use; tested with 12 participants; Best Research Paper Award, IFToMM ISRM 2026.

2. UPPER-LIMB REHABILITATION EXOSKELETON

Bachelor's capstone, 2018 - 2019 | *tested prototype*

- **My role:** Led the design and build.
- **CAD work:** CAD design of the device and its parts.
- **Design decision:** Designed parts so they could be machined and assembled in the college workshop with the tools we had, keeping cost low for a rehabilitation device.
- **Result:** Built and tested with users; Best Major Project Award and INR 100,000 TATA Technologies grant.

3. FORCE ESTIMATION ON A 5-BAR PARALLEL ROBOT

PhD, Aalborg University | *validated test setup*

- **My role:** Estimation method and validation.
- **CAD work:** Worked with the robot's linkage geometry (Jacobian) and variable-stiffness actuators.
- **Design decision:** Estimated tip force from the actuators already in the robot instead of adding a force sensor at the tip, saving cost and weight at the end effector.
- **Result:** Within 10% RMS error of a reference force sensor; used to measure human arm impedance.

SKETCHING

Diploma in Fine Arts. I start every design with hand sketches and can provide sketch samples on request.